RICE | BUSINESS
Jones Graduate School of Business

Analyzing Data - SDC and Event Study

PhD Workshop 2026 Fall

Roadmap for the day

From IPO deals in SDC to published results, in four steps.

1 Using SDC

PULL THE IPOS

Pull the IPO deals from SDC Platinum and the stock data from CRSP on WRDS.
workspace.refinitiv.com/web/Apps/SDCPlatinum

2 Linking to CRSP

SCALE UP AND VERIFY

Link each IPO event to CRSP and scale it up to all deals. Verify the matches, with Ritter and the SEC prospectuses in the decision.

3 Event study

METHODOLOGY

Event study methodology: event days, benchmark, expected return model, abnormal returns and event windows.

4 Replications

VALIDATE OUR WORK

Use the data to replicate 2 published results: IPO descriptive statistics (Lowry, Michaely and Volkova 2017) and an event study (Bradley, Jordan and Ritter 2003).

SDC getting the data

The mechanical part: a tour of SDC and getting the IPOs.

1 Tour of SDC

SESSION AND REPORT

Follow the SDC Manual in the guides folder. SDC runs inside LSEG Workspace: a session holds the actions (content set, filters), a report holds the columns to download.

2 Getting IPOs

FILTER AND EXECUTE

New session, Equity Issues, Common Stock. Filter the listing nation (United States, keep blanks) and the IPO flag, add the report, press Execute. The Lowry Michaely Volkova 2017 - Online Appendix - Extraction and Cleaning of IPO Data from SDC Platinum.pdf is a good guide for columns and decisions (old version).

3 Save

KEEP A LOCAL COPY

Save the session and the report as files, so you can load them again later or share them.

4 Tearsheet

GO OVER AN EXAMPLE

Search the 6-digit CUSIP 71910P (Phoenix Motor): SDC has two deals, 3837860002 and 3932242009. Open both tearsheets and compare them side by side.

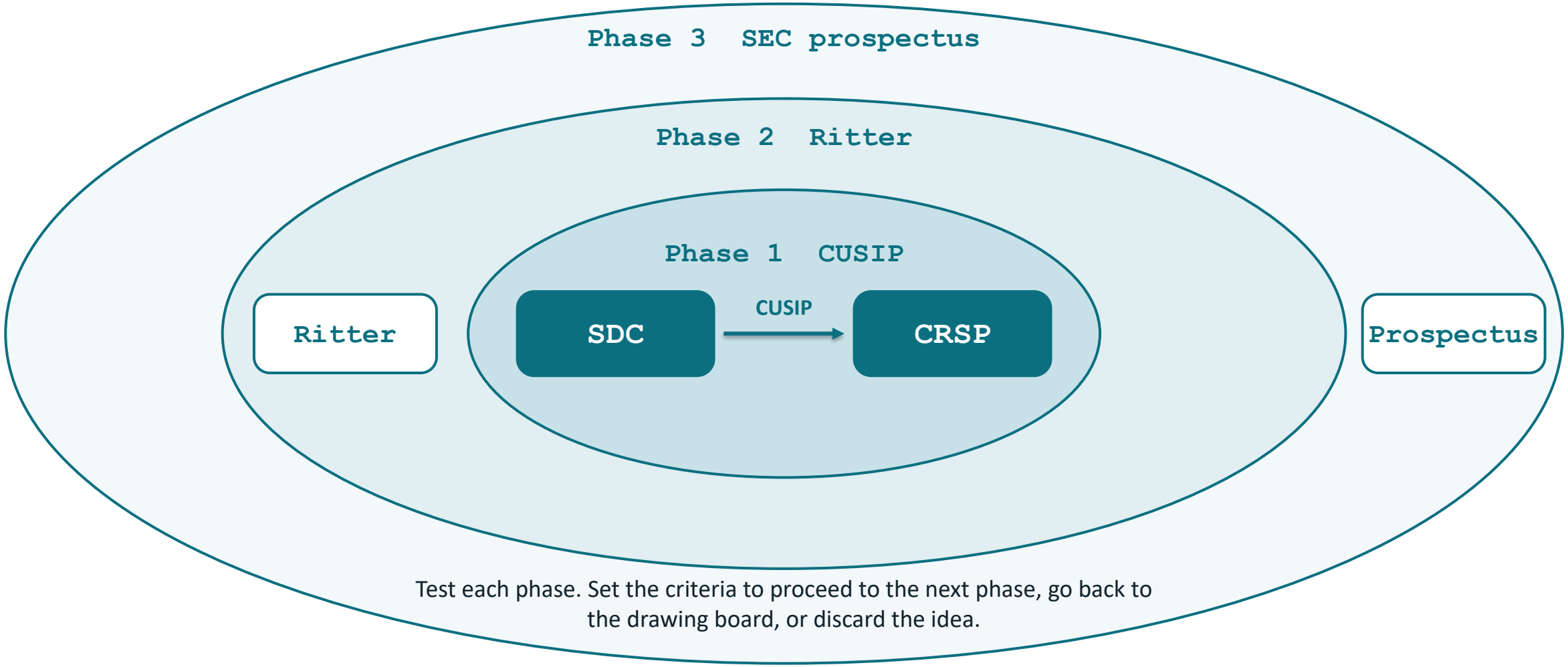
5 Download Excel

EXPORT AS VALUES

Click the Excel icon, Export as values. It fails above about 124k rows (under 120k rows and 20 to 25 columns is safe). Split the pull or patch the page: see the guides folder.

Linking SDC to permno

Each phase adds a source to the decision, from the inside out.



Test each phase. Set the criteria to proceed to the next phase, go back to the drawing board, or discard the idea.

Filters on the SDC download

- Drop a deal when IPO Flag or Original IPO Flag is No
- Same first step as the Lowry, Michaely and Volkova appendix

Filter	Deals left	Dropped
SDC download	59,381	
IPO Flag = Yes	21,198	38,183
Original IPO Flag = Yes	20,715	483
Listing nation United States or blank	20,715	0
Issue date up to the CRSP end (2025-12-31)	20,428	287

Note: the appendix takes 43,817 to 16,454 observations with the same flags (an older download). The SDC session already keeps only United States or blank.

Problems to address, one by one

- 1. The 9-digit CUSIP is missing
- 2. The 6-digit CUSIP differs from the 9-digit
- 3. Two securities of one firm (dual shares)
- 4. The CUSIP fails but the ticker is valid
- 5. Same firm, dates far apart
- 6. The CUSIP matches, but it is another firm

Problem 1: the 9-digit CUSIP is missing

- 5,865 of 20,428 deals have no 9-digit CUSIP
- The 6-digit CUSIP still names the firm

Item	SDC deal 1093848009	CRSP permno 87447
Name	United Parcel Service Inc {UPS}	UNITED PARCEL SERVICE INC
9-digit CUSIP	(missing)	911312106
6-digit CUSIP	911312	911312
Ticker	UPS	UPS
Date	issue date 1999-11-09	first trading day 1999-11-10

Result: pass 3 (6 digits) finds permno 87447, 1 day apart, name score 91.

Problem 2: the 6-digit differs from the 9-digit

- 1,433 of 14,562 deals: 6-digit and 9-digit disagree
- Without a 9-digit, the 6-digit alone can be wrong

Item	SDC deal 3183993002 (shares)	SDC deal 3516135002 (units)
Name	GFL Environmental Inc	GFL Environmental Inc
9-digit CUSIP	36168Q104	36168Q120
6-digit CUSIP field	0C3065	36168Q
First 6 of the 9-digit	36168Q	36168Q
CRSP CUSIP, permno 19291	36168Q104	36168Q104
Pass that found the permno	pass 1 (9 digits)	pass 3 (6 digits)

Result: both deals get permno 19291. The field 0C3065 is not the firm. Of 599 such deals found by the 9-digit, the 6-digit agrees with CRSP in 87.

Problem 3: two securities of one firm

- 92 deals find two or more permnos on 6 digits
- Dual shares: A and B, one SDC deal

Item	SDC deal 1097037009	CRSP permno 83335	CRSP permno 83336
Name	Scania AB	SCANIA AKTIEBOLAG	SCANIA AKTIEBOLAG
9-digit CUSIP	(missing)	806018107	806018206
6-digit CUSIP	806018	806018	806018
Share class	Class A Common Shares	A	B
Date	issue date 1996-04-01	first trading 1996-04-01	first trading 1996-04-01

Result: both start the same day and the share class tells them apart. In 9 of the 92 deals both start within 5 days; in 76 they are over a year apart.

Problem 4: CUSIP fails, ticker is valid

- 287 deals get their permno from the ticker
- SDC has no 9-digit; its 6-digit is not CRSP's

Item	SDC deal 2177338002	CRSP permno 12542
Name	Nielsen Holdings NV	NIELSEN HOLDINGS N V
9-digit CUSIP	(missing)	N63218106, then G6518L108 (2015)
6-digit CUSIP	65410B	N63218, then G6518L
Ticker	NLSN	NLSN
Date	issue date 2011-01-26	first trading day 2011-01-26

Result: same ticker, same day, name score 100: the ticker pass gives permno 12542.

Problem 5: same firm, dates far apart

- 498 deals: same name, dates over a year apart
- Step 1 has no date limit; the gap is saved

Item	SDC deal 517563002	CRSP permno 84181
Name	Canadian National Railway Co	CANADIAN NATIONAL RAILWAY CO
9-digit CUSIP	136375102	136375102
Ticker	CNR	CNI
Date	issue date 1995-11-17	first trading day 1996-12-02

Result: same firm and same CUSIP, but SDC is 381 days before the first CRSP trading day.

Problem 6: CUSIP matches another firm

- SDC's 9-digit CUSIP for Viasystems is Merix's in CRSP
- 32 of 11,967 pass 1 deals have a name score below 50

Item	SDC deal 1093306009	CRSP permno 80553
Name	Viasystems Group Inc	MERIX CORP
9-digit CUSIP	590049102	590049102
6-digit CUSIP field	92553H	(590049, from the 9-digit)
Ticker	VG	MERX
Date	issue date 2000-03-23	first trading day 1994-05-25

Result: all 9 digits match, but the names differ (score 43 of 100) and the dates are 2,129 days apart.

Problem 7: main tranche yes and no

- 1,124 permnos have two deals: one main tranche, one not
- Keep the main tranche

Item	SDC deal 940481002	SDC deal 1093848009
Name	United Parcel Service Inc {UPS}	United Parcel Service Inc {UPS}
Issue date	1999-11-09	1999-11-09
Main tranche	Yes	No
Target market	United States only	UK, Europe, International, Rest of World
CRSP permno	87447	87447

Result: deal 940481002 is kept for permno 87447. Deal 1093848009 is the one from Problem 1 (no 9-digit CUSIP); it is removed.

Problem 8: both deals are main tranche

- 104 permnos have two main tranche deals
- Example: Phoenix Motor, permno 23193, both on 2022-06-07

Item	SDC deal 3837860002	SDC deal 3932242009
Main tranche	Yes	Yes
Offer price	$8.00	$7.50
SDC close at first trade	(blank)	4.059
CRSP close, first trading day	4.06	4.06
Target market	United States only	Rest of World, Asia, Europe
Prospectus status	Incomplete	Completed

Result: step 1 (SDC close equals CRSP close) keeps deal 3932242009, before the US-only rule is used. The final prospectus says $7.50. 002 has S1 and its amendments but no prospectus, the 009 has prospectus information ($8 vs $7.50 offer price).

Summary: SDC to CRSP on Firm Identifiers

- 20,428 SDC IPO deals after initial filters
- Pass 1: all 9 characters (11,967 deals)
- Pass 2: first 8 characters (67 more)
- Pass 3: first 6 characters, the firm (5,464 more)
- Then the ticker (287 more)
- 2,643 deals have no permno yet

Phase 1: Decisions and Rules

- One permno per deal: the first trading day closest to the SDC date
- One deal per permno: keep the main tranche
- Still tied: SDC close equals CRSP close, then US-only target market, then prospectus status Completed, then deal number ending in 002
- A hand fix table covers the rest (38 deals)



- The internet appendix managed these as follows:

Exclude Observations w/o CRSP Record

After the above step, we drop companies for which we were not able to find a proper match. Additionally, we drop observations with duplicated values of `Permno` (usually that happens when a company offers different classes of securities at the same time). These criteria lower the number of observations from 11,103 to 8,995 observations.

Institutional Details

Concept	SDC column	What it tells us (our 20,428 deals)
Offering technique	Offering Technique	Firm Commitment: the underwriters buy the whole issue and resell it (17,598 deals). Others: Best Efforts (1,262), Offer for Subscription (1,409), Placement (760). A deal can list more than one.
Security type	Security Type This Market, Is Unit Issue	Common Stock 13,916; Units 2,260 (shares with warrants); Ordinary Or Common Shares 1,357; Class A Common Shares 1,003. The unit flag is True for only 1,386 deals, so read the security type text.
Tranches	Is Main Tranche, Number of Tranches in Package	A package is the whole offering, a tranche is one part of it (for example the international part). 17,261 deals are a single main tranche. In 2-tranche packages 1,300 deals are main and 1,293 are not.

Institutional Details (continued)

Concept	SDC column	What it tells us (our 20,428 deals)
Target market	Target Market	Where the shares are offered: United States only in 17,411 deals, Canada in 1,243, the rest list regions (UK, Continental Europe, International, Rest Of The World).
Last 3 digits of the deal number	SDC Deal Number	002: 17,405 deals, 17,404 with a US-only target market. 009: 1,782 deals and 091: 1,170 deals, none US-only. Same issuer can have a 002 deal and a 009 deal.
SEC forms	SEC File Form (SDC) and the EDGAR filing list	S-1: the registration statement filed before the IPO (SDC form S1: 8,685 deals). S-1/A: its amendments. 424B4: the final prospectus after pricing (5,742 of our 7,669 prospectuses; 424B1: 1,079). Example: Phoenix Motor, S-1 filed 2021-11-29, 424B4 filed 2022-06-09.

Name matching: a score from 0 to 100

- Take every CRSP name of the permno (not "Last Known", drop a trailing NEW)
- Make variants: as written, with "The", and the words in other orders (4 words or fewer)
- Clean: lowercase, remove (text), "and", punctuation, suffixes like Inc and Corp, spaces
- Score: the mean of six similarity measures
- Keep the best score over the names and variants

Name matching: CRSP flipped the name

- CRSP sometimes writes the last name first
- A name with the words in another order is tried too

Item	SDC deal 519761002	CRSP permno 82581
Name	Henry Schein Inc	SCHEIN HENRY INC
Cleaned as written	henryschein	scheinhenry
CRSP name, words reordered		HENRY SCHEIN INC
Cleaned, reordered		henryschein

Result: score 60.73 as written, 100.00 with the CRSP name reordered. The 9-digit CUSIP already matched; the name score confirms it. The same happens for Steven Madden and Kenneth Cole (20 SDC deals in pass 1).

Name matching: Last Known and NEW

- Last Known: CRSP tags the final name of a permno, e.g. NIELSEN HOLDINGS PLC (Last Known)
  - We skip them: the same name exists untagged, so no name is lost (27,672 of 40,518 permnos have one)

- NEW: CRSP adds it to a successor's name, e.g. C D W CORP NEW (764 names)
  - We drop NEW. It mattered for CDW, AVX and DTM (66.4 became 100) and New Holland (66.9 to 79.2)

Phase 2: Adding Ritter

- SDC stays the base: one Ritter row, or none, for each deal
- Find the row by permno, then CUSIP, then ticker, then name
- Dates within 30 days and a name score of at least 70
- The ticker, or the CUSIP, confirms when the name fails
- 15,203 of 16,536 deals get a row: 14,466 by permno, 368 CUSIP, 274 ticker, 95 name
- Ritter has 16,030 rows: 15,203 are on a deal, 399 match no SDC deal, 428 are on deals dropped for no permno

Phase 2: choosing the permno

- CRSP and Ritter can give different permnos
- Both need a name score of 70 or more; the higher score wins, a tie goes to the closer date

Item	SDC deal 2595328002	CRSP permno 77783	Ritter permno 14990
Name	Anchor Bancorp Wisconsin Inc	ANCHOR BANCORP WISCONSIN INC	Anchor BanCorp Wisconsin Inc
Name score		100	100
Date	issue date 2014-10-21	first trading 8,132 days before	Ritter date 2014-10-22

Result: both score 100, so the closer date wins and the deal gets permno 14990 (Ritter wins). The CRSP permno is an older Anchor Bancorp Wisconsin listing.
All deals: agree 14,466, CRSP only 1,919, Ritter only 123, Ritter wins 21, CRSP wins 6.

Phase 2: exceptions

- Two deals on one Ritter row: the better way keeps it, then the deal number ending in 002 (93 deals lost their row)
- Ritter row wrong: a hand fix table (1 row)
- Ritter information wrong: a hand fix table (11 rows); the row stays but its permno or CUSIP is not used
- No permno after this step: the deal is dropped (2,553 deals)

Swapped in Ritter	Aerovate Therapeutics	Enact Holdings
Ritter IPO date	2021-06-30	2021-09-16
Permno Ritter gives	21906 (CRSP: Enact Holdings)	21518 (CRSP: Aerovate/Jade Biosciences)
Right permno	21518	21906

The recommended permno: rules

- Candidates: the CRSP permno and the permno on the Ritter row
- Same permno: accepted, even with a low name score (132 deals under 70)
- Different: both need a name score of 70 or more; the higher score wins, a tie goes to the closer date
- Neither passes: no permno, and the deal is dropped
- Result: agree 14,466, CRSP only 1,919, Ritter only 123, Ritter wins 21, CRSP wins 6

The recommended permno: sources disagree

- Three ways the two sources can end

Item	Carolina Group	Anchor Bancorp Wisconsin	Stevanato Group
CRSP permno, name score	89303 LOEWS CORP, 42.8	77783 ANCHOR BANCORP WISCONSIN INC, 100	21694 STEVANATO GROUP S P A, 93.4
Ritter permno, name score	89303 LOEWS CORP, 42.8	14990 ANCHOR BANCORP WISCONSIN INC NEW, 100	21775 TEXAS COMMUNITY BANCSHARES INC, 46.2
Rule	same permno: accepted, no test	both 100: the closer date wins	Ritter's is under 70: CRSP's wins
Recommended permno	**89303 [agree]**	**14990 [ritter wins]**	**21694 [crsp wins]**

Result: Carolina Group is a tracking stock of Loews, so the names differ and the sources agree. Stevanato: Ritter's permno belongs to another firm.

Phase 3: adding the prospectus

- Index first: submissions.zip lists the filings of all companies
- Find each deal's filing by SDC file number or company name, within 7 days
- Pick one: 424B4 first, then 424B1, 424B2, 424B5, 424B3
- Download the documents: 9,795 requests, 9,612 worked, 182 not found
- 7,669 deals have a prospectus; documents start with deals from 1997

Phase 3: downloading from the SEC

Rule	What we do	SEC page
Header	Say who you are: the SEC asks for a name and a contact email in the request header	sec.gov/os/accessing-edgar-data
Rate	The SEC maximum is 10 requests per second. We send 2 per second, one at a time, never in parallel	sec.gov/about/webmaster-frequently-asked-questions
No bots or crawling	Ask only for the exact document addresses: no search pages, no following links	sec.gov/os/accessing-edgar-data and sec.gov/robots.txt
Bulk file	One download of submissions.zip (about 1.5 GB), not one request per company	sec.gov/search-filings/edgar-application-programming-interfaces
Ask only for what you need	Save the first 1 MB, never ask twice, stop at the first 403 or 429	sec.gov/os/accessing-edgar-data

Care: the SEC manages requests outside its policy. Our program sends nothing unless download is switched on (v_download=True).

Phase 3: what we read from the prospectus

Item	Found (of 7,669 deals)	Same as SDC (differ)
Offer price	7,572	7,512 (60)
Total shares	7,613	7,393 (220)
Primary shares	7,413	7,119 (294)
Secondary shares	4,959	2,883 (136)
Ticker symbol	7,256	6,528 (489)
Exchange	7,264	not compared
Prospectus date	6,637	compared in days
US shares, international shares, CUSIP	30, 30 and 8	not compared

How: a program reads the first 60,000 characters of each document. A hand fix list (348 deals) replaces values the reader got wrong. The prospectus is an independent source, so a difference with SDC stays in the table.

The recommended date: rules

- Three dates: SDC, Ritter and the prospectus. Two must agree (within 3 days)
- Two of the three agree: that date (SDC first)
- Otherwise the CRSP first trading day is the second source
- All differ: the date closest to CRSP, within 30 days of it
- No CRSP date, or none within 30 days: the SDC date
- Result: it differs from the SDC date in 22 of 16,536 deals

The recommended date: dates disagree

- Two sources within 3 days decide
- The odd one out loses

Item	Atlantic Coast Financial	VISX Inc
SDC date	2011-02-03	1988-10-19
Ritter date	(no Ritter row)	1989-05-11
Prospectus date	2010-11-12	(none before 1997)
CRSP first trading day	2011-02-04 (hand fixed)	1989-05-12
Recommended date	**2011-02-03 (SDC and CRSP)**	**1989-05-11 (Ritter and CRSP)**

Result: Atlantic Coast: the prospectus date is 83 days earlier and loses to SDC and CRSP. VISX: the SDC date is 204 days earlier and loses to Ritter and CRSP.

Which permno and date to use

- Replication 1: the recommendation, permno and date
- Replication 2: the recommended permno; the quiet period starts at the earlier of the recommended date and the CRSP first trading day

You can choose	Permno column	Date column
CRSP, with the hand fixes	`crsp_permno`	`crsp_first_trading_date`
Ritter, with the hand fixes	`ritter_permno (wrong ones flagged)`	`ritter_ipo_date`
The recommendation	**`recommended_permno`**	**`recommended_date`**

Event study: the parameters

Parameter	What it sets	Quiet period run
Event date, day 0	Day 0 is the first trading day on or after the event date	25 days after the IPO
Event window	First and last day around day 0 that the run calculates	days -5 to +119: -5 to +5 are reported, +20 to +119 give the variance
Estimation window	Days used to fit the model (not used by the market-adjusted model)	not used by ma (program setting: -268 to -14)
Model	Expected return: ma, capm, ff3, ff4, ff5 or ff6	ma (market-adjusted)
Benchmark	The market return: mkt, crspvw, crspew or nasdaq, and the _excess version of each	nasdaq
Return type	firm, or firm_excess (the return minus the risk-free rate)	firm

Eventus default: the market model on the CRSP equal-weighted index, fitted on 255 trading days that end 46 days before the event date.

Event study: the risk-free rate

- Excess return: the return minus the risk-free rate (rf)
- Market-adjusted: rf cancels if both sides are raw or both are excess
- In a regression, one raw side and one excess side puts the average rf into the intercept
- Our program: you pick the stock side and the market side separately
- Quiet period run: raw returns on both sides

Risk-free rate: WRDS code, Eventus, ours

Case	Firm return	Market return
WRDS free code: market-adjusted	raw	raw (mktrf + rf)
WRDS free code: market model and Fama-French	raw	minus rf (mktrf)
Eventus 8: two-step, with an estimation window	raw, rf not deducted	raw, rf not deducted
Eventus 8: calendar-time Fama-French (its default)	minus rf	minus rf
Eventus 9: options RfAdjust and NoRfAdjust	you choose	you choose
Our program	firm or firm_excess	raw or _excess

Sources: Eventus 8 Users Guide (PDF pages 34 and 88) and Eventus 9 New and Enhanced Features (pages 7 and 14); the WRDS Python event study code (class EventStudy). Without an Eventus subscription, follow its documentation by hand.
WRDS code: wrds-www.wharton.upenn.edu/pages/wrds-research/applications/python-replications/programming-python-wrds-event-study/

Replication 1: Lowry, Michaely and Volkova, Table 1

Year	IPOs, ours	IPOs, paper	Initial return %, ours	Initial return %, paper	Avg proceeds $M, ours	Avg proceeds $M, paper
1980	74	69	16.4	16.0	32.2	33.5
1985	221	221	8.0	8.0	41.9	42.3
1990	111	118	10.9	11.0	56.4	52.7
1995	464	495	21.3	21.0	78.3	74.3
2000	377	377	55.6	56.0	200.2	198.2
2005	170	167	10.1	10.0	205.8	206.8
2010	109	104	8.8	9.0	323.0	335.0
2016	84	57	12.9	10.0	171.0	184.9
All years	**8,847**	**8,995**	**17.2**	**17.0**	**119.1**	**115.2**

Years 1973-2016: IPO count equal in 8 of 44 years and within 5% in 30; initial return within 1 point in 36. Proceeds are in 2016 dollars, shares offered in this market (median gap 1.8%, all markets 3.6%).
Choices: made by Claude; the sample steps follow the Online Appendix of the paper.

Replication 2: quiet period, Table I

Variable	n	Mean	Std	Min	Max
Offer amount ($M)	1,954 / 1,611	107.13 / 71.60	364.49 / 145.30	4.50 / 5.40	10,620.00 / 2,925.60
Initial return (%)	1,954 / 1,611	36.03 / 37.30	65.70 / 65.60	-43.27 / -43.30	697.50 / 697.50
Managing underwriters	1,954 / 1,611	3.13 / 2.90	1.96 / 1.08	1.00 / 1.00	30.00 / 14.00
Lead underwriter share (%)	1,954 / 1,611	6.81 / 4.84	8.99 / 5.95	0.01 / 0.00	41.22 / 24.50

Each cell: ours / published (Bradley, Jordan and Ritter 2003).
Choices: made by Claude. The instructor chose two: the CRSP share codes (10 and 11, common stock of US companies) and the IPO date, which is the earlier of the CRSP first trading day and the recommended date. Claude found the Nasdaq Composite in CRSP (index 1000503, price return only).

Replication 2: quiet period, Table II

Event day	Avg abnormal return %, ours	paper	t-stat, ours	paper	Gen. sign z, ours	paper
Day -5	0.48	0.66	2.81	3.29	3.52	3.88
Day -4	0.50	0.49	2.92	2.45	2.16	2.23
Day -3	0.54	0.57	3.12	2.85	3.39	3.48
Day -2	0.87	0.83	5.09	4.17	5.20	4.08
Day -1	1.34	1.50	7.83	7.52	7.11	6.88
Day 0	0.64	0.62	3.72	3.10	2.39	1.48
Day 1	0.20	0.26	1.14	1.29	0.80	0.48
Day 2	-0.10	-0.07	-0.61	-0.37	-1.78	-1.27
Day 3	-0.14	-0.22	-0.79	-1.11	-1.28	-1.92
Day 4	-0.22	-0.27	-1.26	-1.33	-0.92	-1.22
Day 5	-0.43	-0.43	-2.50	-2.16	-2.10	-2.67
Days (-2,+2)	2.95	3.12	7.68	7.03	10.82	9.52
Days (-2,-1)	2.22	2.32	9.13	8.26	9.79	8.52
Days (0,+2)	0.73	0.80	2.45	2.33	3.27	2.38

Sample: 1,954 IPOs ours, 1,611 in the paper. Days -5 to +5, then the windows (-2,+2), (-2,-1) and (0,+2).
Choices: made by Claude, except the share codes and the date rule, chosen by the instructor.

Working with Claude: what went wrong

What went wrong	Example	What to do
Kept bringing closed items back as open	Wrote "not decided" for three deals that were already settled	Keep a decisions file and have Claude read it first
Had to be reminded of decisions, or of where something was put	Could not say which SDC field opened the tearsheets	Write each decision and its location in a notes file
Downloaded SEC files without approval and without the header	About 823 requests to sec.gov (two runs) after "create it, do not run it"	Say "create it, do not run it"; check the download switch and the header
Did not follow the CLAUDE.md rules	Searched for the Python path again, though CLAUDE.md gives it	Keep the rules short and ask which rule applies
Code was not optimized; sessions were spent deciding whether and how to parallelize, despite instructions	The SEC zip reader used 13.5 GB of memory in the workers, 2.3 GB after the fix	Give Claude your parallel blueprint and say to follow it
Asked about one item, it answered about the whole sample	You asked about Phoenix only and it went on to other pairs	Say "this case only"